

PIONEER JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATION 2015

**Candidate
Name**

--

Civics

--	--	--	--	--

Index Number

--	--	--	--

H2 BIOLOGY

9648 / 02

17 Sept 2015

2 hours

READ THESE INSTRUCTIONS FIRST

Write your name, class and index number on all the work you hand in.

Write in dark blue or black pen.

You may use a soft pencil for any diagrams, graphs or rough working.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Read the instructions on the answer sheet very carefully.

This document consists of 2 sections:

Section A:

Answer **ALL** questions.

Answers are to be written in the spaces provided.

Section B:

Answer **ONE** question.

Write your answers on the separate writing papers provided.

For Examiner's Use	
Structured Questions	
1	/ 12
2	/ 12
3	/ 12
4	/ 12
5	/ 10
6	/ 12
7	/ 10
8 or 9	/ 20
Total	/ 100

Do not open this booklet until you are told to do so.

Section A: Structured Questions [80 marks]

Answer all the questions in the space provided.

Question 1 [12 marks]

Collagen is the main structural protein of the various connective tissues in animals. As the main component of connective tissue, it is the most abundant protein in mammal, making up from 25% to 35% of the whole-body protein content. **Fig. 1.1** shows the structure of a collagen fibre and collagen fibrils.

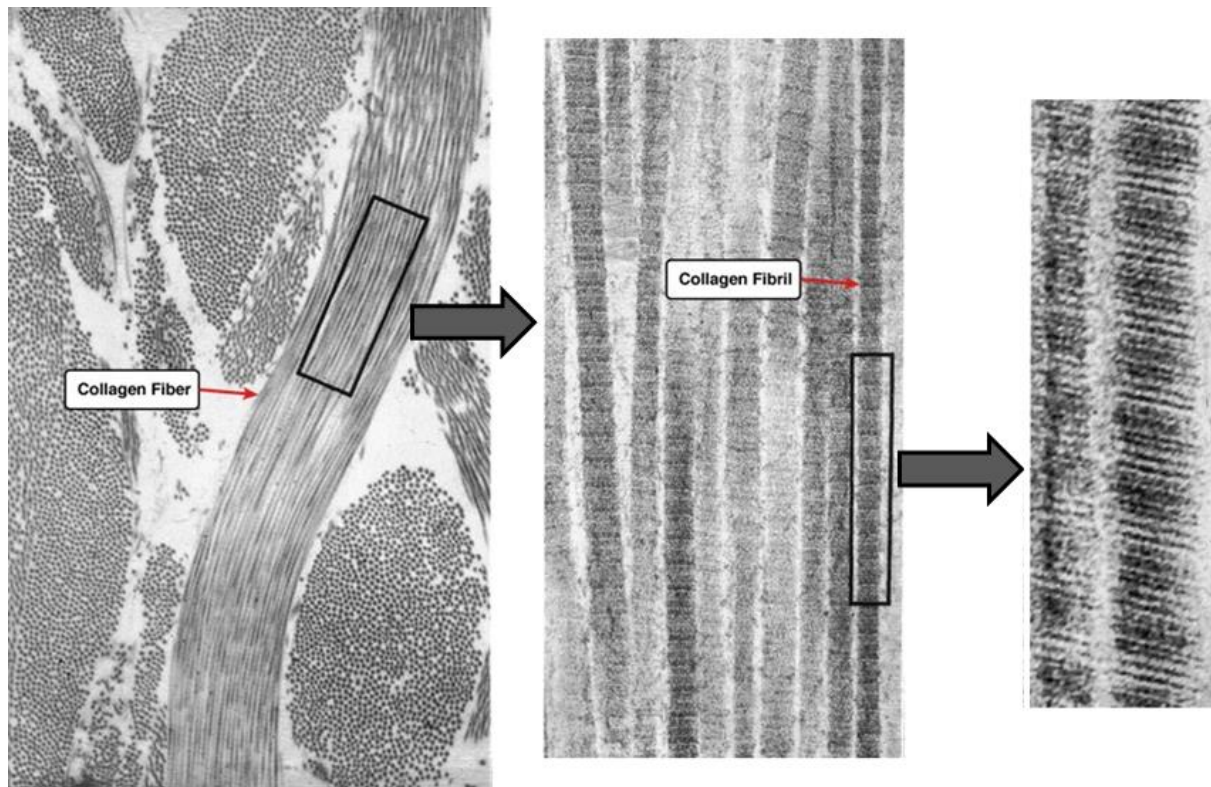


Fig. 1.1

(a) (i) Describe the primary structure of a collagen polypeptide. [2]

- a. Definition of primary structure is the number, type and sequence of amino**
- b. The primary structure of collagen is rich and repetitive in amino acid proline and glycine**

(a) (ii) Explain why the collagen fibril appears as a banded structure. [1]

- a. Staggered arrangement of tropocollagen**
- b. Formation of covalent cross-linkages between tropocollagen**

Besides structural proteins in the human body, there are other proteins that play important roles in cellular processes too. An example would be pepsin, which digests proteins ingested so that amino acids can be taken up by cell for further protein synthesis. **Fig. 1.2** shows the structure of human pepsin.

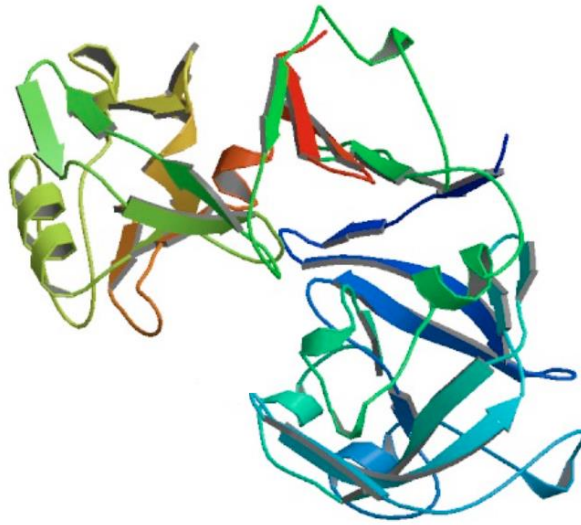


Fig. 1.2

- (a) With reference to **Fig. 1.1** and **Fig. 1.2**, describe two ways how the structure of a collagen fibre differs from that of an enzyme such as pepsin. [2]

Points of comparison

- a. Overall structure**
- b. Number of polypeptide chains involved**
- c. Level of structure**

Any 2 points of comparison

Fig. 1.3 shows the molecular structure of a dipeptide, which is formed from two amino acids.

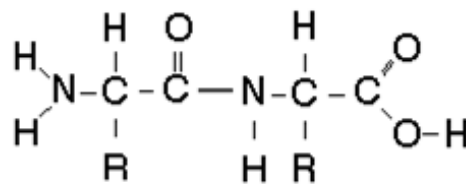


Fig. 1.3

- (c) Describe how the bond holding the two amino acids together may be broken to release the two amino acids. [2]

- a. Describe hydrolysis of peptide bond**
- b. Describe how $-\text{COOH}$ and $-\text{NH}_2$ groups reform to give 2 amino acids**

Another example of proteins with a crucial role in maintaining stable intracellular conditions would be transport proteins embedded in the cell surface membrane (**Fig. 1.4**).

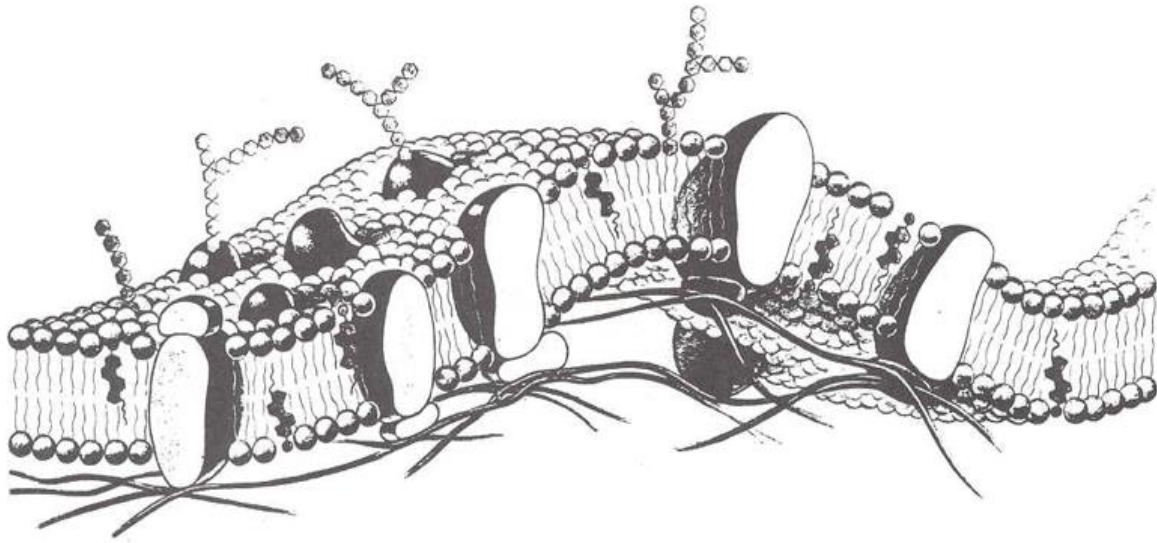


Fig. 1.4

(d) Describe the properties of the phospholipid bilayer. [2]

- a. Fluid mosaic model**
- b. Describe fluidity of bilayer**
- c. Describe proteins embedded in bilayer -> conferring selective permeability**

(e) Explain the significance of transport proteins in found in the membrane. [3]

- a. Charged molecules or ions, polar molecules will be repelled by the hydrophobic core of the membrane**
- b. Transport proteins provide a hydrophilic channel for such molecules**
- c. Describe the different types of transport across the membrane and the transport proteins involved**

Question 2 [12 marks]

Fig. 2.1 below shows DNA replication in an organism.

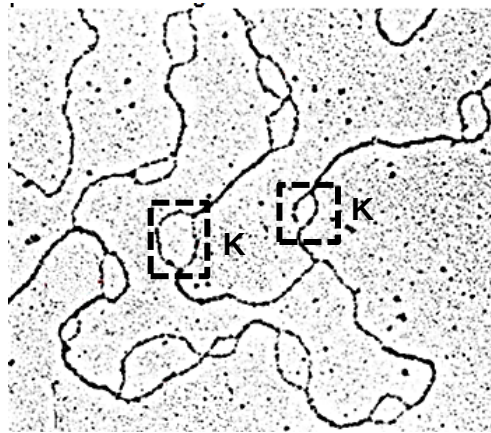


Fig. 2.1

- (a) (i) With reference to **Fig. 2.1**, state, with a reason, if this organism is eukaryotic or prokaryotic. [1]

Eukaryotic as there is more than one origin of replication (multiple replication bubbles)

- (a) (ii) Explain the role of structures **K** in the cell. [1]

a. Speed up the rate of DNA replication

b. Eukaryotic genome is larger than prokaryotic genome

- (b) (i) The DNA replication at each replication fork is sometimes described as '*asymmetrical*' replication as there are differences in the way the daughter strands are being synthesized. State two such differences. [2]

Points of comparison

a. Synthesis of leading and lagging strand – continuous or discontinuous

b. Direction of synthesis of daughter strands with respect to replication fork

c. Number of primers involved in the synthesis of the daughter strands

d. Involvement of ligase

Any 2 points

- (b) (ii) Suggest **two** reasons for these differences in (b) (i). [2]

a. DNA synthesis is always from 5' to 3' -> DNA Pol can only add nucleotides to the free 3' hydroxyl end;

b. DNA is anti-parallel;;

Telomeres undergo characteristic length changes over time in normal somatic and germline cells, and in premature ageing syndromes. In contrast to germ cells, which have high telomerase activity (indicated by the plus symbol on the graph), most somatic cells possess low or absent telomerase activity (indicated by the minus symbol on the graph). Telomere shortening is thought to contribute to the onset of degenerative diseases that occur during human ageing, eg. heart failure; immunosenescence (infections); digestive tract atrophies (ulcers); infertility; reduced viability of stem cells; reduced angiogenic potential; reduced wound-healing; and loss of body mass. In addition, several human premature ageing syndromes show an accelerated rate of telomere shortening, therefore resulting in an early onset of ageing-related pathologies. Examples are Ataxia telangiectasia (*ATM*); Werner syndrome (*WRN*); Bloom syndrome (*BLM*), etc.

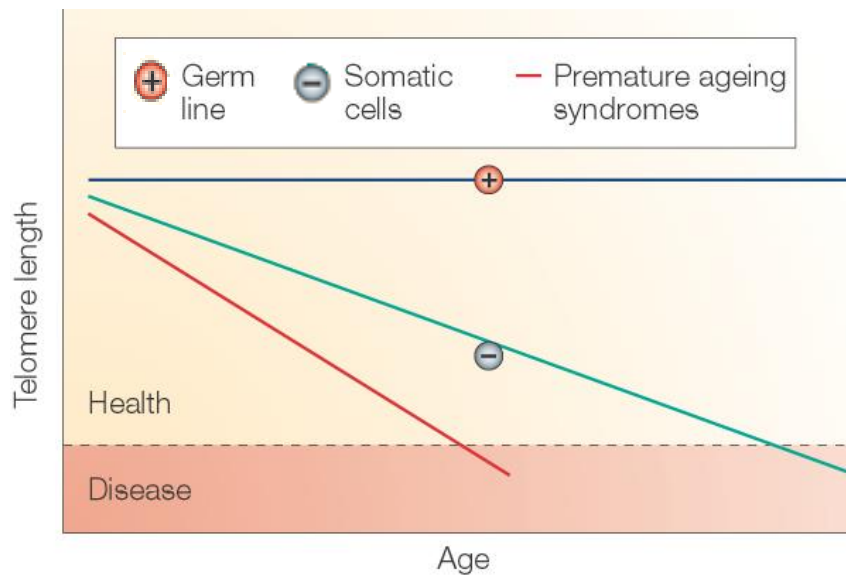


Fig. 2.2

(d) With reference to **Fig. 2.2**,

(i) explain the role of telomeres. [2]

Role:

- a. To provide stability to the chromosome by rendering chromosome ends generally inert in interactions with other chromosome ends
- b. acts as a buffer of genes near the ends of chromosomes
- c. To prevent genomic instability and development of cancer in human aged cells by limiting the number of cell divisions

(ii) explain what happens when telomeres reach a critical length. [2]

- a. The progressive telomere loss eventually leads to critically short telomeres -> triggers cell arrest and apoptosis
- b. Describe on senescence and onset of degenerative diseases

(iii) account for the differences in telomere lengths in germ cells and somatic cells. [2]

a. Quote trends from the graph above

b. Explanation with respect to presence of telomerase and end-replication problem

Question 3 [12 marks]

Fragile X Syndrome (FXS), the most common inherited form of mental retardation and autism, which is caused by inactivation of the *Fmr1* gene, which encodes the Fragile X Mental Retardation Protein (FMRP). In a normal person, FMRP regulates the synthesis of synaptic proteins. **Fig 3.1** shows how patients with FXS have non-functional FMRP, resulting in accelerated synaptic protein synthesis that leads to abnormal synaptic function and intellectual disability.

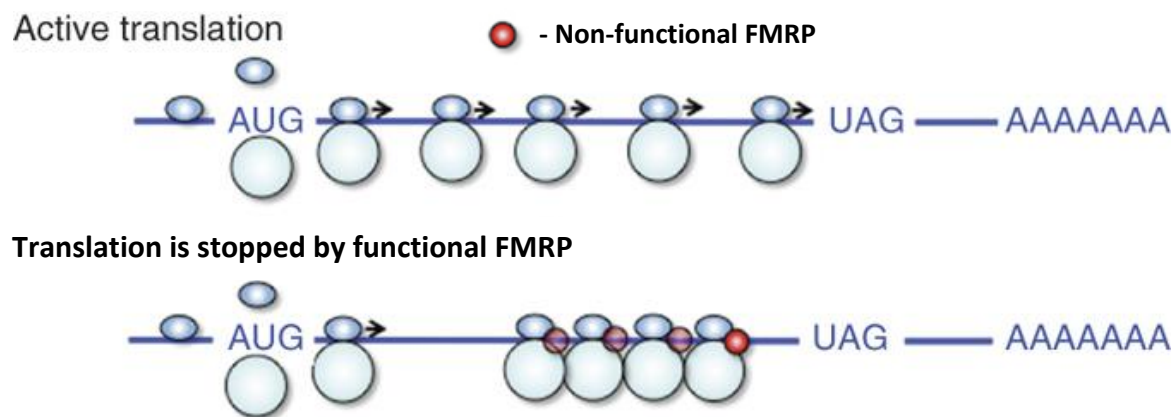


Fig. 3.1

With reference to **Fig. 3.1**,

(a) (i) state the level of control of gene expression and role exerted by FMRP. [1]

Translational Control;;

(ii) state the role played by FMRP. [1]

FMRP: Repressor of translation

(iii) suggest how FMRP may regulate gene expression at the level stated in (a)(i) in a normal person. [2]

a. Describe how FMRP binds to mRNA

b. Ribosome is blocked -> translation elongation is inhibited

(iv) suggest the advantage of this level of control over that of an earlier stage of gene expression. [1]

- a. **Ref. to faster formation of protein; ensures a faster response in protein synthesis;; OR**
- b. **Faster to maintain translation homeostasis to cope with rapid changes in protein concentrations;;**

Excessive synthesis of synaptic proteins, which causes FXS is due to inactivation of FMRP. The sequences of both normal and mutated FMRP are reflected in the figure below.

Functional FMRP	His – arg – leu – thr – pro – glu – ser –lys.....
Mutated FMRP	His – arg – leu – thr – pro – val – ser –lys.....

Fig. 3.2

(b) With reference to **Fig. 3.2**, explain how a mutation in the FMRP could result in increased protein synthesis that causes Fragile X syndrome. [3]

- a. **Point mutation results via base-pair substitution**
- b. **Missense mutation -> quote different amino acid coded for**
- c. **Effect of new amino acid on the 3D structure of protein resulting in increased ribosomal translation**

(c) In a particular mutation, one of the bases in the FMR1 gene sequence was replaced by another base. However, the FMR1 protein remained functional. Suggest two reasons how this may be possible. [2]

- a. **Silent mutation**
- b. **Same amino acid coded for due to the degeneracy of the genetic code**
- c. **Missense mutation**
- d. **Change in triplet code -> Different amino acid coded for**
- e. **Same properties in R group of amino acid/ changed properties but not in critical region of protein**
- f. **Substitution in the intron**
- g. **As introns will be excise during post-transcriptional modification**

Any two valid points + elaboration

(d) Describe two ways in which accuracy in translation is achieved. [2]

- a. **Specificity of amino acyl trna-synthetase in amino acid activation**
- b. **Complementary base pairing between codons on mrna and anticodon on trna**

Question 4 [12 marks]

The following shows an electron micrograph of a bacteria cell infected by foreign particles.

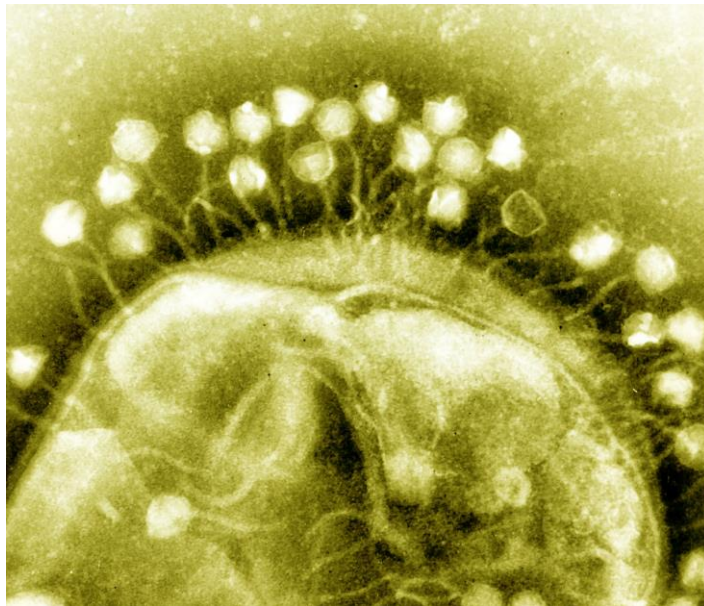


Fig. 4.1

The image was enlarged and the electron micrograph of a single particle is shown below.

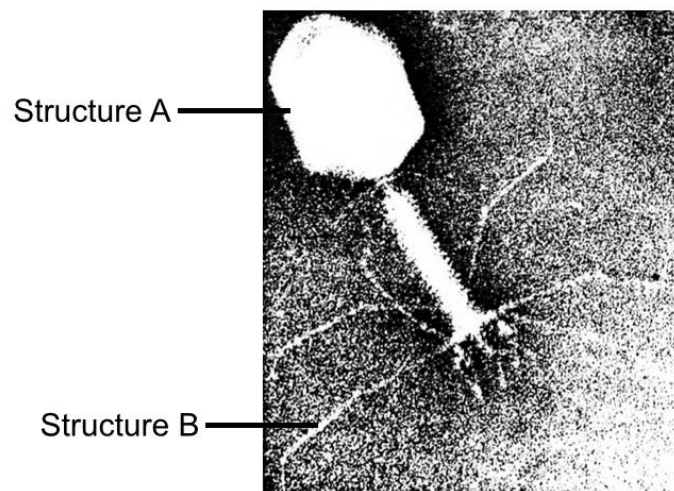


Fig. 4.2

(a) Identify this foreign particle. [1]

T4 bacteriophage;;

(b) Identify the structural components labeled on **Fig. 4.2**. [1]

- **Structure A: Icosahedral nucleocapsid**
- **Structure B: Tail Fibres**

(c) Describe how the foreign particle enters the bacteria as shown in **Fig. 4.1**. [2]

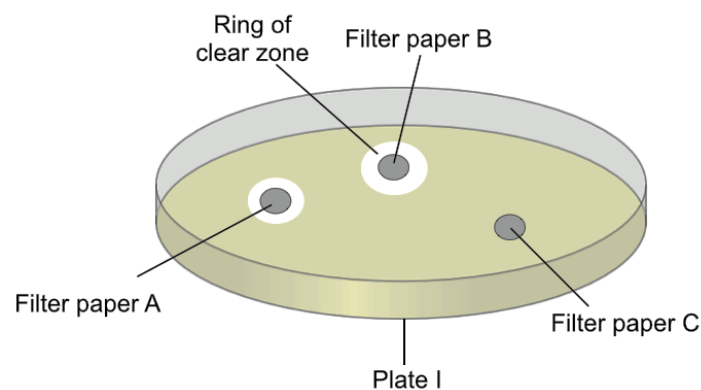
- a. T4 phage uses tail fibres to recognise and adsorb to receptor sites on the host bacterial cell wall;;
- b. Its tail fibers anchor the base plate while the tail sheath contracts;;
- c. The viral nucleic acid/double stranded DNA is injected into the bacterial cytoplasm;;

(d) Outline the function of **structure A**. [3]

- a. The capsid will protect the viral ds DNA
- b. contains genes that code for the viral components eg;;
- c. genes that codes for early and late proteins (provide examples)
- d. template for replication of viral nucleic acid;

The following experiment was carried out to investigate the effect of bacteriophage and antibiotic on *E. coli*.

Three circular filter papers were soaked with either bacteriophage or antibiotics and placed on an agar plate, labeled Plate I, with a lawn of *E. coli*. The plate with the filter paper was then incubated for 24 hours and rings of clear zones were observed (**Fig. 4.3a**). These zones represent bacteria killed.



The same experiment was repeated on a second agar plate containing a different strain of *E. coli* (**Fig. 4.3b**).

Filter Paper A	Ampicillin
Filter Paper B	T4 bacteriophage
Filter Paper C	Lambda phage

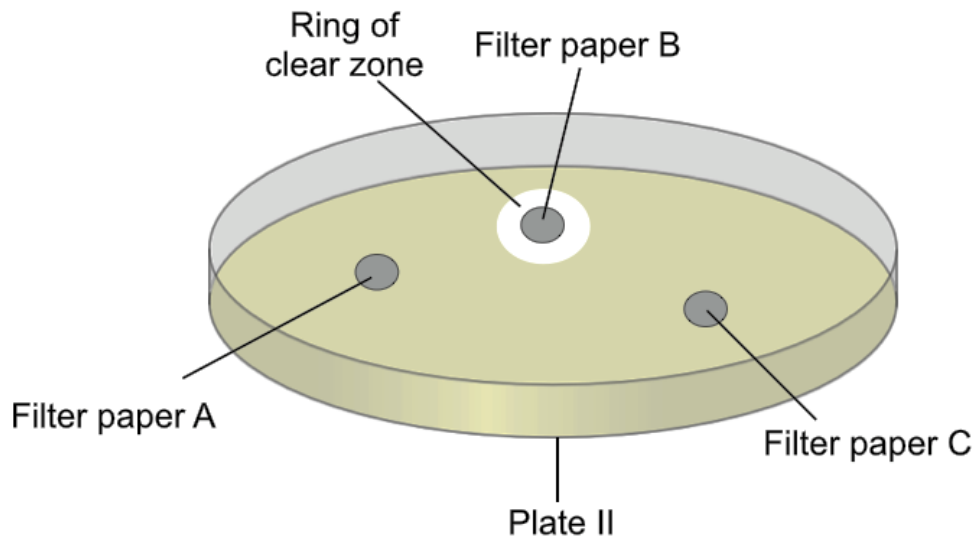


Fig. 4.3b

(e) With reference to **Fig. 4.3a and 4.3b**,

(i) identify the plate that will contain ampicillin resistant *E. coli*. Explain your answer. [1]

a. Plate II;

b. no clear zones observed around filter paper A -> bacteria can survive

(ii) account for the differences observed for filter papers B and C after 24 hours of incubation. [3]

a. (Quote) In both plates, there was a ring of clear zone when T4 bacteriophage was present whereas no ring of clear zone was present when lambda phage was used

b. T4 undergoes a lytic cycle -> results in lysis of the E.coli upon release of new virions (hence explains the clearing)

c. Lambda phages undergo lysogenic cycle where viral genome is integrated into E.coli's genome as a prophage -> no immediate lysis of E.coli

Phage therapy is the therapeutic use of bacteriophages to treat pathogenic bacterial infections. Phage therapy can be very effective in certain conditions and has some unique advantages over antibiotics.

(f) Suggest which of the two bacteriophages – T4 phage or lambda phage– will be more suitable for use in phage therapy. Explain your answer. [1]

a. T4

b. Effectively brings down the number of bacteria causing the infection.

Question 5 [10 marks]

In cattle, coats may be solid white, solid brown, or beige. When true breeding solid whites are mated with true-breeding solid brown, the F1 generation consists of all solid white individuals. Matings among the F1 generation have resulted in the following ratio:

23 solid white
6 beige
3 solid brown

(a) Draw a genetic diagram of the above described cross. [4]

Let W represent white colour fur
Let w represent coloured fur
Let B represent beige fur
Let bb represent brown fur

F1 Phenotype F1 genotype Meiosis	white WwBb		X	white WwBb	
Gametes	WB Wb wB wb			WB Wb wB wb	
Offspring genotype and phenotype		WB	Wb	wB	wb
	WB	WWBB white	WWBb white	WwBB white	WwBb White
	Wb	WWBb white	WWbb white	WwBb white	Wwbb white
	wB	WwBB white	WwBb white	wwBB beige	wwBb beige
	wb	WwBb white	Wwbb white	wwBb beige	wwbb brown
Offspring phenotypic ratio	12 white : 3 beige : 1 brown				

(b) Explain the genetic basis of this type of inheritance. [2]

- Dominant epistasis -> W is epistatic to B and/or b alleles.
- Dominant A codes for e.g. enzyme that catalyses breakdown of coloured pigments
- Dominant B codes for enzyme that catalyses formation of beige pigments

The chi-squared (X^2) test was performed on the F2 data giving a calculated value for X^2 of 4.73.

Degrees of freedom	Probability, p				
	0.10	0.05	0.02	0.01	0.001
1	2.71	3.84	5.41	6.64	10.83
2	4.61	5.99	7.82	9.21	13.82
3	6.25	7.82	9.84	11.35	16.27
4	7.78	9.49	11.67	13.28	18.47

(c) State what conclusion may be drawn regarding the observed F2 data. [2]

- a. Since the p value of $\chi^2_{\text{calculated}}$ is between $P=0.05$ and 0.1 at $df=2$, which is $> p=0.05$ of χ^2_{critical} ,
b. there is no significant difference between observed and expected ratio of 12:3:1
c. any difference between observed and expected ratio is due to random chance

In a particular variety of tomato plant, the allele for red fruit colour (**A**) is dominant over the allele for orange fruit (**a**) and the allele for green base when ripe (**B**) is dominant over the allele for no green base when ripe (**b**).

Two students, Eugene and Cornelius crossed plant with red fruit and green base when ripe with pure bred plant with orange fruit and no green bases when ripe. The phenotypes of 50 offspring of each of Eugene's and Cornelius's crosses were recorded and are shown in **Table 5.1**.

	Phenotypes of offspring of test crosses			
	Red fruit with green base	Red fruit with no green base	Orange fruit with green base	Orange fruit with no green base
Eugene's cross	23	4	3	20
Cornelius's cross	3	21	23	3

(d) Using the symbols given, state the genotypes of the tomato plants in Eugene's and Cornelius's crosses. [2]

Eugene's cross: $\frac{AB}{ab} \times \frac{ab}{ab}$ *Reject: different symbols used*
Cornelius's cross: $\frac{Ab}{aB} \times \frac{ab}{ab}$

Question 6 [12 marks]

Ras is a GTP-binding protein that helps to broadcast signals from the cell surface to other parts of the cell to stimulate cell proliferation. If Ras function is inhibited, the cell proliferation responses normally induced by the activated receptors do not occur. Conversely, if a hyperactive mutant, Ras protein, is introduced into some cell lines, the effect on cell proliferation is the same as that induced by the binding of ligands to cell surface receptors. This hyperactive product of mutant ras gene promotes the development of cancer. **Fig. 6.1** below shows the sequence of events in a Ras cell cycle-stimulating pathway.

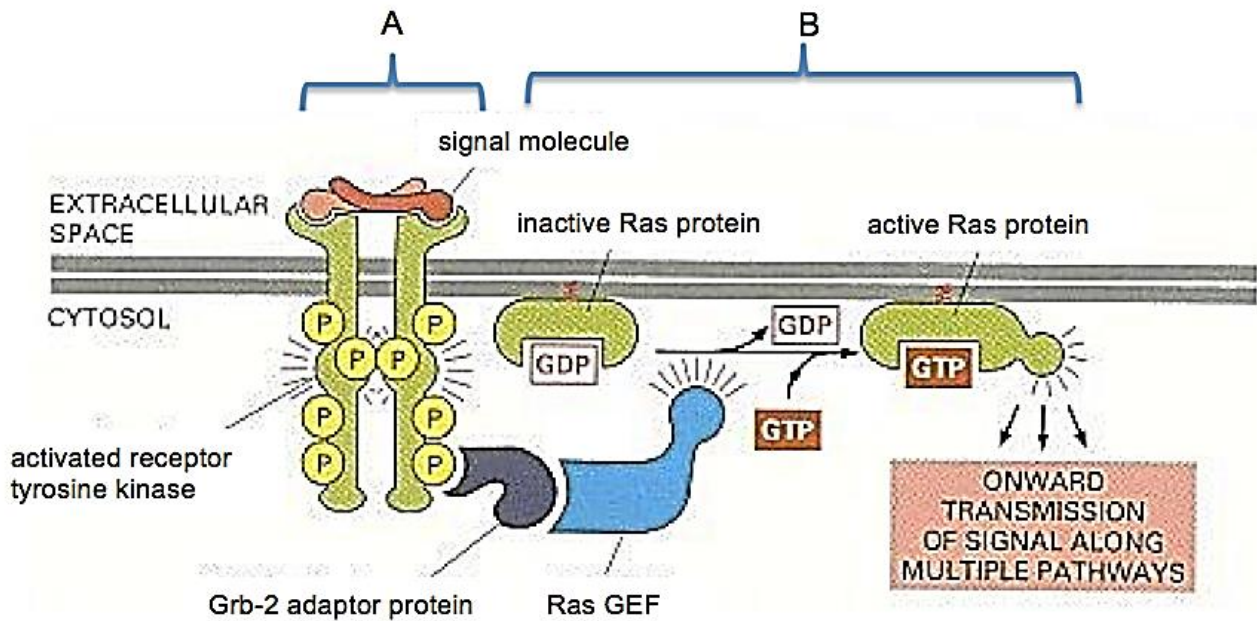


Fig. 6.1

(a) With reference to Fig. 6.1,

(i) briefly describe the stages A and B. [4]

A

- a. **Dimerisation of RTK**
- b. **Cross-phosphorylation of tyrosine amino acids on each other's tyrosine amino acid tails followed by autophosphorylation**

B

- c. **relay protein(s), Grb-2 adaptor protein recognizes and binds to specific phosphorylated tyrosine(s) which acts as docking protein**
- d. **Activation of Ras GEF and Ras protein**
- e. **to relay signal by activating other proteins or enzymes to bring about various cellular responses**

(ii) observe that Ras protein is a plasma membrane bound G-protein. Suggest how the activated G-protein can relay the signal from the ligand to activate cellular responses. [2]

- a. **By activating enzymes such as adenylyl cyclase which will catalyse production of cAMP**
- b. **Activation of phosphorylation cascade for cellular responses**

The receptor for epidermal growth factor (EGF) can be activated by a deletion that removes part of its extracellular domain. These mutant receptors are able to form active dimers even in the absence of EGF, and thus they produce a stimulatory signal inappropriately.

Besides mutation to the gene that codes for the extracellular domains of the receptors, 30% of human tumors have a hyperactive Ras mutation.

(b) Suggest how mutation in the Ras gene may bring about cancer. [3]

- a. Ras-gene is a proto-oncogene -> gain-of-function mutation results in oncogene
- b. Hyperactive Ras protein that cannot hydrolysed bound GTP due to mutated GTPase domain
- c. Can no longer be turned off even when ligand/growth factor is no longer bound to the receptor tyrosine complexes
- d. Activate other cellular proteins which stimulate cell division continuously → tumor

(c) Briefly explain why the development of cancer is a multi-step process. [3]

- a. There are several regulatory checkpoints to prevent a cell from progressing into a cancerous one.
- b. Multiple gene loci for TSGs and proto-oncogenes need to be mutated -> Describe gain-of-function mutation and loss-of-function mutation for proto-oncogene and TSG respectively
- c. Gene for telomerase is activated
- d. Tumour cells becomes locally invasive
- e. Additional mutations required for metastasis - the entry into bloodstream or lymphatic systems, travel via circulation and colonization of remote sites.

Question 7 [10 marks]

Soil salinity is one of the major environmental stresses causing a significant decrease in productivity in world culture agriculture as it can inhibit the growth of plants. Salinity refers to the occurrence of salts in ground water. Ions contributing to the salinity include Na^+ , K^+ , Mg^{2+} and Cl^- .

There are a few species of plants of genus *Arabidopsis* that are closely related to each other showing differences in ability to transport these ions out of the cell. They have all arisen from a common ancestor, but have evolved to live in different soil conditions.

The molecular basis for variation in the type of ions being transported out of the cell appears to be due to HKT gene. HKT gene codes for an ion channel known to transport these salts out of the cell (see **Fig. 7.1**). Some HKT only transport a specific ion while others can transport different ions simultaneously (see **Table 7.2**).

Based on the molecular information of all genes found in *Arabidopsis*, it has been studied that a common ancestor of *Arabidopsis* plants, *A.dentata*, contains HKT1 which codes for a protein that transports only Mg^{2+} . (see **Fig. 7.3**)

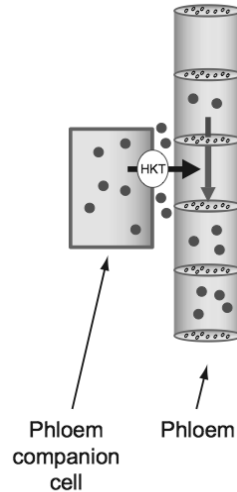


Fig. 7.1

Table 7.1

Salt tolerance of <i>Arabidopsis</i> plants	Type of soil salinity that it cannot survive in	Types of HKT present
High tolerance	Able to survive in all types of soil salinity	HKT 3 - transports only Na^+ , K^+ and Cl^-
Moderate tolerance	Soil rich in magnesium and potassium	HKT 2 – transports only Cl^- and Na^+
Sensitive	Soil rich in sodium, potassium and chloride	HKT 1 – transports only Mg^{2+}

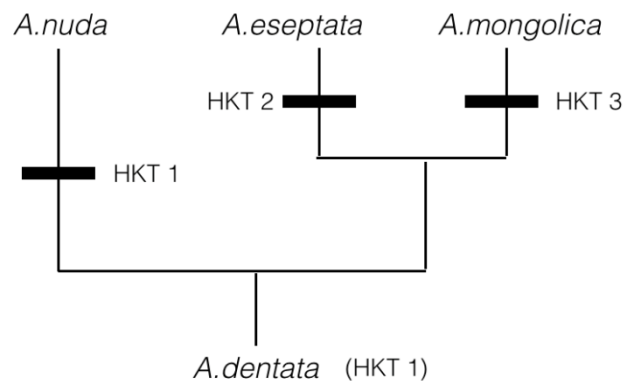


Fig. 7.2

(a) With reference to **Table 7.1** and **Fig. 7.2**,

(i) state the environments that *A. nuda*, *A. eseptata* and *A. mongolica* are likely to have evolved in. [1]

- a. *A.nuda*: **environment with only magnesium ions**
- b. *A.eseptata*: **environment with sodium and chloride ions**
- c. *A.mongolica*: **environment with sodium, chloride and potasisum ions**

(ii) explain how the molecular evidence described supports Darwin's theory of natural selection. [4]

- a. HKT gene also found in common ancestor, *A.dentata* of the various *Arabidopsis* plants;
- b. Existing variation in selectivity of HKT could be due to mutations -> giving rise to different HKT;
- c. Selection pressure: types of ions present in the soil
- d. Depending on type of ions in the ground in which the plants grow in, those that are tolerant to those ions will have differential survival and reproductive rate.
- e. These plants will pass on the favourable HKT allele to the next generation of offspring -> allele frequency increases in gene pool

(iii) identify two pairs of *Arabidopsis* species that each shows close relatedness. Explain your answer. [2]

- a. 1st pair: *A.esptata* and *A.mongolica*;
- b. They diverged from a more recent common ancestor;
- c. 2nd pair: *A.dentata* and *A.nuda*;
- d. No divergence observed for this gene as they both have the HKT allele that codes for HKT1;

In a separate study on other Eukaryotes, neutral theory was put forth in accounting for much of the evolutionary changes that occur at the molecular level. The molecular clock for fibrinopeptides (peptides that aid in blood clotting), hemoglobin, cytochrome c and histone H4 was constructed to study the evolutionary rates of these proteins (see Fig. 7.3).

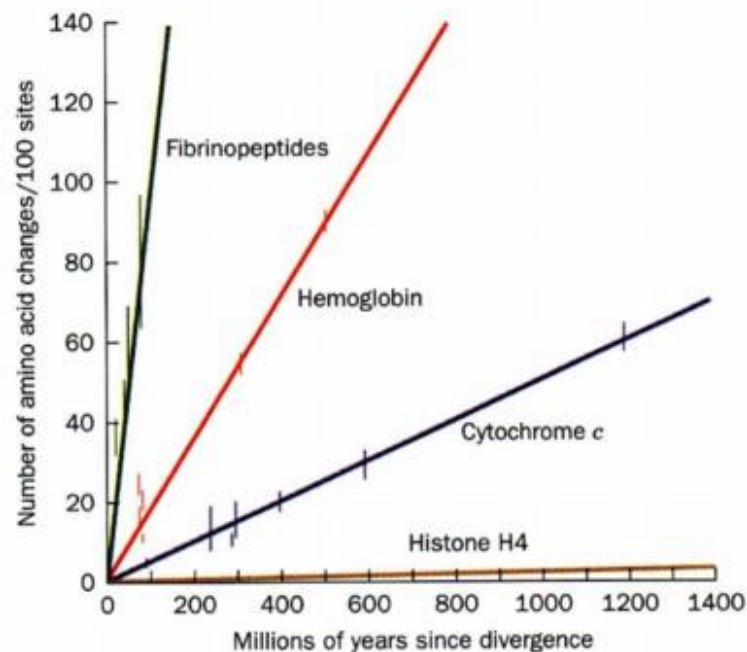


Fig. 7.3

(b) Explain the basis of a molecular clock with respect cytochrome c. [1]

- a. Based on the assumption that neutral mutations
- b. arise at a constant rate

(c) Account for the different evolutionary rates between fibrinopeptides and cytochrome c. [2]

- a. Less critical gene evolves faster than important genes (Quote relevant data from graph)
- b. It is assumed that the rate of mutation across all genes coding for the proteins are the same
- c. Effect of the mutation in a critical gene is likely to be more deleterious/harmful to an organism -> only neutral mutations can accumulate

Section B [20 marks]

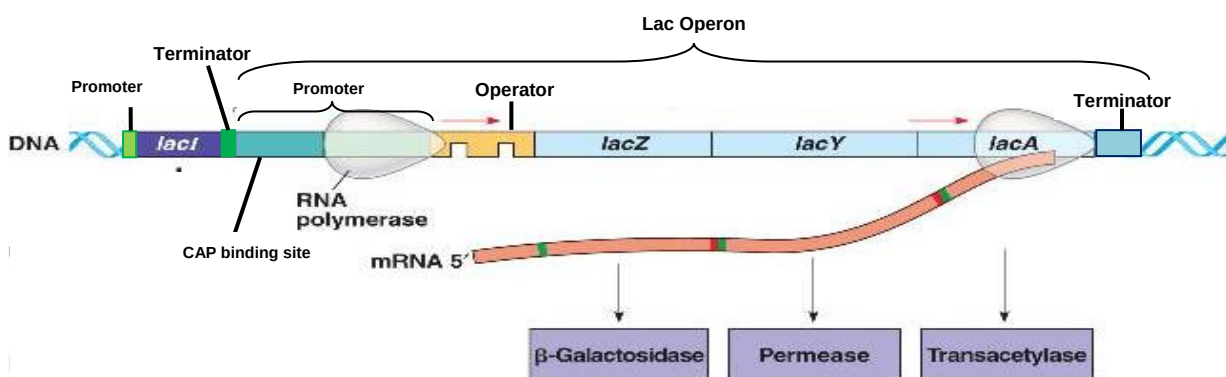
Answer **only ONE** question.

Write your answers on the separate answer paper provided. Your answers should be illustrated by large, clear labeled diagrams, where appropriate. Your answers must be in continuous prose, where appropriate.

Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

Question 8

(a) Describe the structure of a lac operon and explain how it may be regulated. [9]



Structure of operon (max 3 marks):

- a. An operon is a sequence of DNA including a promoter, an operator, one or more structural genes and terminator sequence which are controlled together as a single transcription unit to produce polycistronic mRNA in the process of transcription.
- b. Briefly describe function of the abovementioned sequences

How operon is regulated:

Negative control: (max 3 marks)

- c. When lactose is absent, repressor remains bound to the repressor and transcription of lac operon is switched off;
- d. When lactose is present, some of the lactose gets transported into the cells and are converted into the inducer, allolactose.
- e. The inducer binds to the repressor and alters its conformation so it cannot bind to the operator
- f. RNA polymerase can now transcribe and translate the structural genes into permease, transacetylase and β -galactosidase

Positive control: (max 3 marks)

- g. When glucose and lactose are both present in its environment, prokaryotes preferentially uses glucose
- h. cyclic AMP (cAMP), which accumulates when glucose is in low concentration in the cell -> binds with CAP to form cAMP-CAP complex
- i. This binding of active CAP to CAP-binding site strengthens the affinity of the promoter region for RNA polymerase -> rate of transcription increases

(b) Describe the process and significance of binary fission in bacteria. [4]

- a. Binary fission preceded by replication of the bacterial chromosome.
- b. Two identical chromosomes attaches to the plasma membrane
- c. As cell grows, the plasma membrane between attachment points of chromosomes enlarges and eventually fuses (with cell wall) around the middle of the cells

Significance:

- d. Binary fission is an asexual reproductive process -> most of the bacteria in a colony are genetically identical to the parent cell.
- e. reproduces quickly in a favorable environment

(c) Describe the process and significance of transformation and transduction in bacteria. [7]

Transformation: (max 2)

- a. uptake of *naked, foreign DNA* from the surrounding environment.
- b. cell surface protein on bacteria can recognize and transport DNA from closely related species into the cell
- c. the imported strand crosses over with the homologous region of recipient bacterial chromosomal DNA via homologous recombination

Transduction (max 3)

- d. In lytic cycle, phage enzymes hydrolyze the bacterial chromosome into small pieces which accidentally gets packaged into capsid during assembly of virions
- e. Defective phage is released and can infect another bacterium
- f. This foreign bacterial DNA can subsequently replace the homologous region of recipient cell's chromosome –homologous recombination

Significance (max 2)

- g. Genetic recombination via genetic exchange in bacteria produces new allelic combinations
- h. Genetic variation results in variations in structure and adaptations
- i. Bacteria may gain antibiotic resistant genes, other xenobiotic resistant genes and genes that have the ability to use a new metabolites

Question 9

(a) Compare and contrast between oxidative phosphorylation and photophosphorylation. [8]

	Oxidative phosphorylation	Photophosphorylation
a. Location	Inner membrane of mitochondrion	Thylakoid membrane of chloroplast
b. Source of energy for synthesis of ATP	oxidation of glucose	Light energy - required for photolysis and excitation of pigment molecules.
c. Electron donors	NADH and FADH ₂	Water is the electron donor in the non-cyclic photophosphorylation while photosystem I is the electron donor in the cyclic pathway.
d. Electron acceptors	Oxygen is the final electron acceptor	NADP ⁺ is the final electron acceptor in the non-cyclic pathway while PSI is the electron acceptor in the cyclic pathway.
e. Establishing proton gradient for synthesis of ATP	Protons are pumped from matrix into the intermembranal space	Protons are pumped from stroma into the thylakoid space
f. Products	ATP and water; NAD ⁺ and FAD regenerated	ATP, NADPH and oxygen produced

g. Components of ETC	Consists of NADH dehydrogenase, FADH2 dehydrogenase and cytochrome oxidase	Consists of plastoquinone, a complex of 2 cytochromes and plastocyanin
Similarities		
i. Energy released from flow of electrons down ETC	Energy released from electron transport due to progressively lower energy level of electrons in electron carriers -> used to create proton gradient	

(b) Describe the structural adaptations of mitochondria for its function in ATP synthesis. [6]

Structure	Function
A1. Highly folded inner membrane/ Cristae	A2. increase surface area for embedment of e^- carriers / stalked particles
B1. Outer and Inner membrane impermeable to H^+	B2. Allows generation and maintenance of a H^+ gradient -> to generate PMF
C1. Membrane is fluid in nature	C2. Allow for movement of electron carriers along the membrane to facilitate the transfer of energized electrons from one electron carrier to the next.
D1. Inner membrane contains a mosaic of transmembrane or integral proteins, such as: D2. ETC → a series electron and proton carriers D5. ATP synthase complex	D3. These proteins carriers transport energized electrons down the ETC -> Energy from transfer of electrons used to pump H^+ into intermembranal space → D4. build up a proton motive force in the form of a H^+ gradient D6. To harness the energy from diffusion of H^+ down electrochemical gradient to the phosphorylation of ADP to ATP- > chemiosmosis
E1. Outer membrane of mitochondrion allows compartmentalisation of different reactions to take place within cell	E2. Separates substrates, enzymes and reaction conditions of Kreb's cycle/oxidative phosphorylation/ ATP synthesis in mitochondria from glycolysis in cytosol

(c) Describe the role of NAD⁺ in aerobic respiration. [6]

- a. NAD⁺ is a coenzyme of the dehydrogenase enzymes and the oxidizing agent for glycolysis, link reaction and Krebs cycle
- b. NAD⁺ are proton and electron carriers -> 2H⁺ and 2e⁻ will be passed to NAD⁺ which become reduced to form NADH + H⁺
- c. Each NADH formed represents stored energy that can be used to make ATP when electrons complete their exergonic fall down an energy gradient from NADH to oxygen.
- d. Describe the number of NAD⁺ used in glycolysis, link reaction and Krebs cycle
- e. Transfers electrons and protons to oxidative phosphorylation for production of ATP →
- f. oxidation of 1 NADH produces 3 ATP
- g. Reoxidized NAD⁺ must return to glycolysis, link rxn & krebs cycle to allow aerobic respiration to continue